

9.8.19

AI25BTECH11017-BALU

Question:

- 1) Two circles $x^2 + y^2 = 6$ and $x^2 + y^2 - 6x + 8 = 0$ are given. Then the equation of the circle through their points of intersection and the point $(1, 1)$ is (1980)
- (a) $x^2 + y^2 - 6x + 4 = 0$
 - (b) $x^2 + y^2 - 3x + 1 = 0$
 - (c) $x^2 + y^2 - 4y + 2 = 0$
 - (d) none of these

Solution:

Let us solve the given equation theoretically and then verify the solution computationally
According to the question,
equation circles are

$$\mathbf{X}^T \mathbf{X} - 6 = 0 \quad (1.1)$$

$$\mathbf{X}^T \mathbf{X} + 2 \begin{pmatrix} -3 & 0 \end{pmatrix} \mathbf{X} + 8 = 0 \quad (1.2)$$

substitute 1.1 in 1.2

$$\begin{pmatrix} -3 & 0 \end{pmatrix} \mathbf{X} = -7 \quad (1.3)$$

$$x = \frac{7}{3} \quad (1.4)$$

substitute 1.4 in 1.1

$$y = \frac{\sqrt{5}}{3}, \frac{-\sqrt{5}}{3} \quad (1.5)$$

let equation of required circle is

$$\mathbf{X}^T \mathbf{X} + 2\mathbf{u}^T \mathbf{X} + f = 0 \quad (1.6)$$

$$\mathbf{u}^T \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{-f}{2} - 1 \quad (1.7)$$

$$\mathbf{u}^T \begin{pmatrix} \frac{7}{3} \\ \frac{\sqrt{5}}{3} \end{pmatrix} = \frac{-f}{2} - 3 \quad (1.8)$$

$$\mathbf{u}^T \begin{pmatrix} \frac{7}{3} \\ \frac{-\sqrt{5}}{3} \end{pmatrix} = \frac{-f}{2} - 3 \quad (1.9)$$

$$\begin{pmatrix} 1 & 1 \\ \frac{7}{3} & \frac{\sqrt{5}}{3} \\ \frac{7}{3} & \frac{-\sqrt{5}}{3} \end{pmatrix} \mathbf{X} = \begin{pmatrix} \frac{-f}{2} - 1 \\ \frac{-f}{2} - 3 \\ \frac{-f}{2} - 3 \end{pmatrix} \quad (1.10)$$

$$\text{rank}\begin{pmatrix} 1 & 1 & \frac{-f}{2}-1 \\ \frac{7}{3} & \frac{\sqrt{5}}{3} & \frac{-f}{2}-3 \\ \frac{7}{3} & \frac{-\sqrt{5}}{3} & \frac{-f}{2}-3 \end{pmatrix} = 2$$

(1.11)

$$f = 1$$

(1.12)

$$\mathbf{u} = \begin{pmatrix} \frac{-3}{2} \\ 0 \end{pmatrix}$$

(1.13)

the required circle equation is

$$x^2 + y^2 - 3x + 1 = 0$$

(1.14)

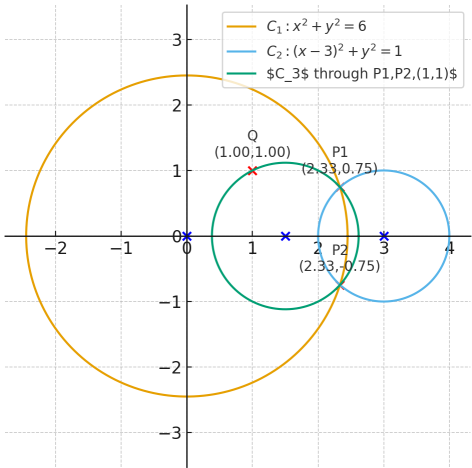


Fig. 1.1